

School of Electrical Engineering and Computer Science,
Washington State University,
Pullman, WA, USA.

Website 🌐: chibuikeugwu.github.io
Email ✉: chibuike.ugwu@wsu.edu

RESEARCH INTERESTS

Trustworthy Machine Learning (TML), large language models (LLM) hallucination mitigation, Conformal Prediction, Deep Learning, Multi-target Regression with Representation Learning, Computer Vision, and Generative AI.

RESEARCH SUMMARY

My general research interests are in **Artificial Intelligence (AI)** and **Machine Learning (ML)**, with a main focus on developing robust, safe, and trustworthy ML algorithms and theoretical frameworks for addressing critical real-world challenges within safety-sensitive domains. My current work focuses on:

- Building trustworthy Large Language Model (LLM) hallucination control policies.
- Adapting calibration-based uncertainty quantification methods, like Conformal Prediction, to facilitate effective and secure human-ML collaboration.
- Developing uncertainty-aware energy management methods for wearable Internet of Things (IoT) devices.

EDUCATION

Ph.D., Computer Science Jul., 2022 – May, 2027
Washington State University
Pullman, WA, USA
Advisors: Prof. Jana Doppa and Prof. Yan Yan
Thesis: Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

M.Sc., Computer Science 🎓 Aug., 2022 – May, 2025
Washington State University
Pullman, WA, USA
Thesis: Conformalized Uncertainty Regions for Machine Learning-Based Multiple Cognitive Health Measures from Smartwatch Sensor Data.

B.Sc., Statistics 🎓 (First Class Honors) Sept., 2014 – Nov., 2018
University of Nigeria
Nsukka, Nigeria
Thesis: Sensitivity of Randomization test and F-test in Some Experimental Designs
Class rank: Best Graduating Student

PROFESSIONAL APPOINTMENTS

Research Assistant Aug 2022 – Present
EECS Department – Washington State University
Pullman, WA, USA
– Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

Teaching Assistant and Guest Lecturer Aug 2022 – Dec 2024
EECS Department – Washington State University
Pullman, WA, USA
– CptS 223: Advanced Data Structures C/C++ (Fall-2022)
– CptS 315: Introduction to Data Mining (Spring-2023)
– CptS 223: Advanced Data Structures C/C++ (Fall-2023)
– CptS 315: Introduction to Data Mining (Spring-2024)
– CptS 223: Advanced Data Structures C/C++ (Fall-2024)

Assistant Lecturer 2019 – 2020
Statistics Department – Kano State Polytechnic
Kano, Nigeria

– Course: Introduction to Statistical Inference

AWARDS AND HONORS

🏆 Nakahara Tsuyoshi and Mary Fellowship (\$2000)	Nov., 2024
🏆 Outstanding Graduate Teaching Assistant in EECS Award <i>Voiland College of Engineering, Washington State University</i>	Apr., 2024
🏆 EducationUSA Opportunity Fund Program (OFP) Scholar. <i>USA Embassy, Abuja, Nigeria</i>	Jun., 2022
🏆 Best Graduating Student <i>Statistics Department – University of Nigeria</i>	Nov., 2018
🏆 Winner: Nigerian Statistical Association (NSA) competition <i>Representative of the University of Nigeria.</i>	May, 2018
🏆 Gold medalist: NSA competition individual category <i>Representative of the University of Nigeria.</i>	May, 2018

PROJECTS/SOFTWARE

Adaptive Prediction Region (APR) [code] – Developed a novel framework (APR) for constructing trustworthy, adaptive uncertainty regions for multi-target regression tasks.	Aug., 2024
ML Adversary: Attack and Defense (Course Project) [code] – Designed and implemented a web-based application for dog breed classification, incorporating defenses against adversarial attacks to enhance model robustness and reliability.	Nov., 2023
Multi-target Conformal Prediction (MTCP) [code] – Designed software (MTCP) to predict calibrated joint regions from independent targets in multi-target regression tasks.	Nov., 2023

PUBLICATIONS

1. [\[IJCAI'25\]](#) Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Sustainable Wearables for Health Applications and Beyond via Uncertainty-Aware Energy Management.** *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, 2025 (* denotes equal contribution)
2. [\[DAC'25\]](#) Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Uncertainty-Aware Energy Management for Wearable IoT Devices with Conformal Prediction.** *Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC)*, 2025. (* denotes equal contribution)
3. [\[TECS'25\]](#) Pratyush Dhingra, Chibuike E. Ugwu, Janardhan Rao Doppa, and Partha Pratim Pande. **ERGo: Energy-Efficient Hybrid Graph Neural Network Training on Heterogeneous Processing-In-Memory Architecture.** *ACM Transactions on Embedded Computing Systems (TECS)*, 2025.
4. [\[QREI'23\]](#) Abimibola V. Oladugba, Chibuike E. Ugwu, and Uchenna C. Onwuamaeze. **Sensitivity and robustness of randomization test and F-test in some experimental designs.** *Quality and Reliability Engineering International*, 39(7), 2967-2974, 2023.

PAPERS UNDER REVIEW

1. [AAAI'26] Chibuike E. Ugwu, Roschelle Fritz, Diane J. Cook, and Janardhan Doppa. **Clinician-in-the-Loop Smart Home System to Detect Urinary Tract Infection Flare-Ups via Uncertainty-Aware Decision Support.** *Proceedings of 40th AAAI Conference on Artificial Intelligence (AAAI)* 2026.
2. [AAAI'26] Hooman Shahrokhi, Chibuike E. Ugwu, Yan Yan, and Janardhan Doppa. **Controlling Coverage for Batch Evaluation via Vanilla Conformal Prediction.** *Proceedings of 40th AAAI Conference on Artificial Intelligence (AAAI)* 2026.
3. [TMLR'25] Chibuike E. Ugwu, Yan Yan, Diane J. Cook, and Janardhan Doppa. **Uncertainty Regions for Multi-Target Regression via Input-Dependent Conformal Calibration.** *Transactions of Machine Learning Research (TMLR)*, 2025.
4. [HEALTH'25] Chibuike E. Ugwu, Yan Yan, Diane J. Cook, Maureen Schmitter-Edgecombe, and Janardhan Doppa. **Importance-Weighted Calibrated Region Prediction of Multi-target Cognitive Health Measures from Smartwatch Sensor Data.** *ACM Transactions on Computing for Healthcare*, 2025.
5. [TODAES'25] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Trading Off Performance and Sustainability in Internet of Things: An Uncertainty-Aware Hierarchical Energy Management Approach.** *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2025. (* denotes equal contribution)

PROFESSIONAL SERVICES AND OUTREACH ACTIVITIES

PROGRAM COMMITTEE

1. Association for the Advancement of Artificial Intelligence (AAAI), 2026.
2. International Conference of Machine Learning (ICML), 2025.
3. AAAI Conference on Artificial Intelligence (AAAI), 2025.
4. AAAI Conference on Artificial Intelligence (AAAI), AI for Social Impact Track, 2025.

TECHNICAL/PROFESSIONAL EVENTS

1. Volunteer for International Joint Conference on Artificial Intelligence (IJCAI), 2025.
2. Mentor and Judge for Digital AgAthon (AgAID Institute), 2025.
3. Instructor for WSU Summer Programming Camp for Middle Schoolers, 2025.